

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 14

10 July 2025

Overview:

Today's session introduced us to Logistic Regression, a classification algorithm that helps predict categorical outcomes — especially binary results like Yes/No or True/False. Unlike linear regression, which deals with continuous values, logistic regression is used when the outcome is discrete.

Learning Objectives:

- Understand the purpose of logistic regression in classification
- Learn the difference between linear and logistic regression
- Understand the Sigmoid Function and its role in prediction
- Explore how the model calculates probability and converts it into binary output

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression which predicts continuous values it predicts the probability that an input belongs to a specific class. It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1. It uses sigmoid function to convert inputs into a probability value between 0 and 1.

Types of Logistic Regression

Logistic regression can be classified into three main types based on the nature of the dependent variable:

1. **Binomial Logistic Regression:** This type is used when the dependent variable has only two possible categories. Examples include Yes/No, Pass/Fail or 0/1.
2. **Multinomial Logistic Regression:** This is used when the dependent variable has three or more possible categories that are not ordered. For example, classifying animals into categories like "cat," "dog" or "sheep."
3. **Ordinal Logistic Regression:** This type applies when the dependent variable has three or more categories with a natural order or ranking. Examples include ratings like "low," "medium" and "high."

Sigmoid Function:

This function converts any real-valued number into a value between 0 and 1:

$$S(x) = 1 / (1 + e^{-x})$$

If the output is > 0.5 , we predict Class 1, otherwise Class 0.

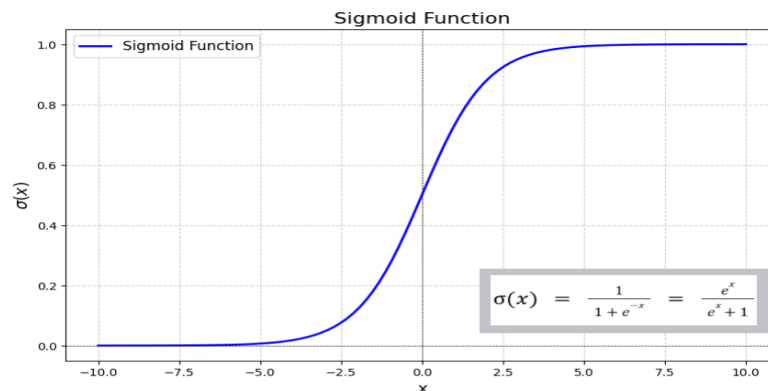


Fig. Sigmoid Function

Properties of the Sigmoid Function

The sigmoid function has several key properties that make it a popular choice in machine learning and neural networks:

1. **Domain:** The domain of the sigmoid function is all real numbers. This means that you can input any real number into the sigmoid function, and it will produce a valid output.
2. **Asymptotes:** As x approaches positive infinity, $\sigma(x)$ approaches 1. Conversely, as x approaches negative infinity, $\sigma(x)$ approaches 0. This property ensures that the function never actually reaches 0 or 1 but gets arbitrarily close.
3. **Monotonicity:** The sigmoid function is monotonically increasing, meaning that as the input increases, the output also increases.
4. **Differentiability:** The sigmoid function is differentiable, which allows for the calculation of gradients during the training of machine learning models.

Application

- Email Spam Detection (Spam / Not Spam)
- Disease Diagnosis (Positive / Negative)
- Customer Churn Prediction (Leave / Stay)

Conclusion:

Logistic regression helped us explore how machines make decisions in classification problems. The introduction of the sigmoid function and understanding how the algorithm handles probabilities made the session very insightful. We now know how machine learning can be used to answer “yes or no” type questions with data-driven confidence.